Cell Structure and the Cell Membrane

All living cells are enclosed by a cell membrane, a selectively permeable phospholipid bilayer studded with embedded proteins. The membrane controls which substances enter and leave the cell, using passive diffusion for small nonpolar molecules and membrane transport proteins (channels and pumps) for ions and large or polar molecules.

Eukaryotic cells additionally contain a nucleus, which houses the cell's DNA, and membrane-bound organelles such as the endoplasmic reticulum and Golgi apparatus, which modify and transport proteins.

Mitochondria and Cellular Respiration

Mitochondria are the organelles primarily responsible for producing ATP, the cell's main energy currency, through cellular respiration. Cellular respiration breaks down glucose in the presence of oxygen through glycolysis, the citric acid (Krebs) cycle, and the electron transport chain, yielding far more ATP per glucose molecule than fermentation alone.

Mitochondria have their own circular DNA and a double membrane; the inner membrane is folded into cristae, which increase the surface area available for the electron transport chain.

Photosynthesis

Photosynthesis is the process by which plants, algae, and some bacteria convert light energy into chemical energy stored in glucose, using carbon dioxide and water as inputs and releasing oxygen as a byproduct. It occurs in chloroplasts and has two stages: the light-dependent reactions, which occur in the thylakoid membrane and produce ATP and NADPH, and the light-independent reactions (the Calvin cycle), which occur in the stroma and use that ATP and NADPH to fix carbon dioxide into glucose.

Photosynthesis and cellular respiration are, in a simplified sense, chemically complementary processes: one stores energy from light into glucose, and the other releases that stored energy from glucose.

DNA Structure and Replication

DNA is a double helix made of two complementary strands of nucleotides, held together by hydrogen bonds between base pairs: adenine pairs with thymine, and guanine pairs with cytosine. This complementary base pairing is the basis for accurate DNA replication.

During replication, the enzyme helicase unwinds the double helix, and DNA polymerase synthesizes a new complementary strand from each original strand, producing two identical double-stranded DNA molecules. Because each new molecule retains one original strand and gains one new strand, this mechanism is called semiconservative replication.

Mitosis

Mitosis is the process by which a single eukaryotic cell divides to produce two genetically identical daughter cells, each with the same number of chromosomes as the parent cell. It proceeds through prophase (chromosomes condense), metaphase (chromosomes align at the cell's equator), anaphase (sister chromatids separate and move to opposite poles), and telophase (two new nuclei form).

Mitosis is used for growth, tissue repair, and asexual reproduction in many organisms, and it is distinct from meiosis, which produces genetically varied reproductive cells.

Meiosis and Protein Synthesis

Meiosis is a specialized form of cell division that produces four genetically distinct haploid gametes from one diploid cell, through two successive divisions (meiosis I and meiosis II) and a single round of DNA replication. Crossing over during meiosis I, where homologous chromosomes exchange genetic material, is a major source of genetic variation.

Protein synthesis converts genetic information into functional proteins in two stages: transcription, where RNA polymerase copies a gene's DNA sequence into messenger RNA (mRNA) in the nucleus, and translation, where a ribosome reads the mRNA in the cytoplasm and assembles the corresponding chain of amino acids.